Middle Miocene magnetostratigraphy of the Pebas Formation at Tamshiyacu and Santa Carina (Iquitos, Peru): a C5AAr (∼13.2–13.4 Ma) tie for the Pan-Amazonian wetland and its *Purussaurus* record

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Working draft (Nov. 2026). The GPTS correlation below rests on the biostratigraphic exclusion test of Fig. 3; detrital-zircon U–Pb maximum depositional ages (five samples, Arizona LaserChron, in analysis) will provide the independent test and tie these sections to the same U–Pb chronology that calibrates the La Venta age model.

Abstract

The middle Miocene Pebas Formation records the Pan-Amazonian "mega-wetland," and its lignite beds yield abundant vertebrate fossils — yet the timing and tempo of that record remain poorly constrained, with age uncertainties of a million years or more, because the strata are rarely tied directly to the geomagnetic polarity time scale (GPTS). We present paleomagnetic u-channel records from two cutbank sections ∼170 km apart near Iquitos, Peru — Tamshiyacu (TAM) and Santa Carina (SC), the localities of Gingras et al. (2002) — each preserving a single, well-resolved polarity reversal. Reconstruction of core azimuth recovers geographic declination, so the reversal is expressed antipodally in both inclination and declination. Tied by the reversal, both sections share a normal-to-reversed transition — the C5ABn–C5AAr boundary — whereas their lignite flooding surfaces do not align and are demonstrably diachronous. Four convergent constraints — the polarity pattern, a palynological floor (>12.7 Ma), a *Purussaurus* ceiling (<13.8 Ma), and U-Pb detrital-zircon ages from the *Purussaurus neivensis*–bearing La Venta section (youngest zircon populations ∼13.9 Ma; Mora-Rojas et al., 2023) — converge on **chron C5AAr, ∼13.2–13.4 Ma**. A *Purussaurus* cranial horizon discovered at Tamshiyacu in 2025 falls within C5AAr, dating it to a ∼180 kyr window (∼13.2 Ma) — ∼1 Myr older than the dated *P. neivensis* horizon at La Venta (the Monkey Beds, chron C5An.1n, ≤12.17 Ma) — a sub-million-year calibration that extends the dated record of the genus in the Pan-Amazonian wetland and is relevant to evolutionary and biogeographic models. Down-core X-ray fluorescence shows iron is dominantly authigenic, recording reducing, episodically brackish bottom water; net accumulation is very low (∼3 m/Myr), marking a condensed, incomplete succession. These records provide the first direct GPTS calibration of the Pebas at Iquitos and a chronologic bridge to the La Venta fauna.

INTRODUCTION

The Miocene reorganization of northern South America — Andean uplift, the Pebas mega-wetland, and the eventual birth of the transcontinental Amazon — is central to Neotropical biogeography (Hoorn et al., 2010a; Wesselingh & Salo, 2006). The formation's lignites and shell beds yield an abundant vertebrate fauna — crocodylians, turtles, rodents, and the giant caiman *Purussaurus* — recovered almost yearly from actively eroding cutbanks; the Iquitos Pebas alone has produced a hyperdiverse crocodylian community, including the ∼13 Ma gavialoid *Gryposuchus pachakamue* whose skull records parallel evolution in the proto-Amazonian mega-wetland (Salas-Gismondi et al., 2015, 2016). Yet individual fossil horizons typically carry age uncertainties of a million years or more, because the age framework of the Pebas Formation rests largely on molluscan and palynological biostratigraphy and sparse radiometric ages and its strata are seldom tied directly to the GPTS. Such coarse control limits the fauna's use in the evolutionary and biogeographic

models that increasingly demand sub-million-year resolution to test rates of diversification, dispersal, and body-size evolution. Whether Pebas deposition was lacustrine and essentially freshwater (Wesselingh & Salo, 2006) or tidally influenced and periodically brackish-marine (Räsänen et al., 1995; Gingras et al., 2002) has been debated for decades, and the correlation of scattered outcrops to one another and to dated faunas such as La Venta remains loose.

Here we calibrate two Pebas sections near Iquitos, Peru, to the GPTS using paleomagnetic u-channel data, and integrate the result with sedimentology, ichnology (after Gingras et al., 2002), down-core geochemistry, and vertebrate and U-Pb age control. The sections are the same cutbanks — Tamshiyacu and Santa Julia II (here "Santa Carina") — whose ichnology first argued for tidal influence (Gingras et al., 2002), allowing our magnetostratigraphy to be read directly against an established facies and process model.

GEOLOGIC SETTING AND METHODS

The Pebas Formation (≈Solimões Fm. of Brazil) accumulated in the Andean foreland/back-arc of western Amazonia during the middle Miocene, as a low-gradient system of shallow lakes, swamps, and floodplains periodically connected to the sea (Hoorn, 1993; Hoorn et al., 2010a,b). At Tamshiyacu ($4.01°$S, $73.15°$W) and Santa Carina ($3.36°$S, $71.79°$W; Fig. 1) the formation comprises blue-grey laminated and massive muds (facies F1A/F1B of Gingras et al., 2002), heterolithic silty sands (F2A–F2D), rooted lignites (F3A), winnowed mollusc shell beds (F3B), and *Glossifungites*-demarcated firmgrounds (F3C), organized into lignite-capped parasequences.

Oriented u-channels were sampled from continuous mud-dominated intervals and measured at Oregon State University. Characteristic remanent magnetization (ChRM) was isolated from despiked, high-coercivity (30–70 mT) steps by principal-component analysis — the upper bound chosen to exclude a high-field gyroremanent (GRM) contribution in the most pyritic intervals (see *Discussion*); at $\sim$4°S the geocentric axial dipole field is shallow, so normal polarity corresponds to negative inclination. Because the u-channels were azimuthally unoriented, polarity is determined from *inclination sign alone*, which is independent of declination; declinations are treated as relative only, and the reversal is expressed as coupled inclination and $\sim$180° declination changes (a qualitative reversal test). Bed-by-bed lithologic logs were rendered with `pystrat`; parasequences, grading, ichnology and physical structures are transferred from the logs of Gingras et al. (2002) (Tamshiyacu parasequences TmA–TmB; Santa Carina = their Santa Julia II, SJA–SJC) by tying their metres to ours at the shared lignite-capped flooding surfaces. The La Venta comparison uses the composite standard section of Mora-Rojas et al. (2023) (bed-by-bed lithofacies, their supplement s6) and their reinterpretation of the Flynn et al. (1997) polarity sequence (supplement s15), with reversals tied to the ATNTS2012 (Hilgen et al., 2012). Down-core geochemistry was measured by energy-dispersive X-ray fluorescence (XRF) core scanning. Decompacted accumulation rates conserve the solid fraction, restoring depositional thickness as $h_0 = h\,(1-\phi)/(1-\phi_0)$ (Sclater & Christie, 1980): we take an initial porosity $\phi_0 = 0.70$–0.80, as measured in modern and Pleistocene subaqueous muds (Meade, 1966) and in the near-surface Pleistocene sediment of the ODP reference sites used in Fig. 4 (Mix et al., 2003), and a present porosity $\phi = 0.30$–0.45 appropriate to lithified mudstone at the Pebas burial depths (Sclater & Christie, 1980), giving a decompaction factor of $\sim$2–3. Compacted (measured) rates are reported alongside the decompacted values.

RESULTS

Polarity records. Each section preserves shallow, dual-polarity ChRM directions and a single robust reversal. Tamshiyacu is normal-below, reversed-above; Santa Carina is normal–reversed–normal. At each

reversal the inclination changes sign in step with a $\sim$180° swing in relative declination (a qualitative reversal check; see Methods), consistent with a primary (detrital to early-diagenetic) remanence. At Tamshiyacu the uppermost core group (the *Purussaurus*-bearing lignite and capping beds) is NRM-only with poor susceptibility and yields no interpretable direction; usable polarity is therefore restricted to the lower $\sim$2.5 m, and the upper $\sim$1.2 m is shown as no data, although XRF was scanned over the full $\sim$3.8 m (Fig. 2).

Shared reversal. Hung on the reversal, both sections show the same normal-to-reversed transition at the datum — the C5ABn–C5AAr boundary — with Santa Carina continuing up into the succeeding normal chron (C5AAn) and Tamshiyacu truncated within C5AAr (Fig. 2). The lignite flooding surfaces do not coincide once the reversals are matched, demonstrating that they are diachronous facies boundaries.

Facies and parasequences. Both sections resolve into the lignite-capped, coarsening-upward bay-margin parasequences of Gingras et al. (2002), each founded on a *Glossifungites* (*Thalassinoides*) firmground (Fig. 2). Tamshiyacu spans TmA — laminated mud coarsening into sand–mud interbeds, capped by a hummocky, storm-emplaced sand that fines upward into bioturbated silty sand and a laterally continuous ironstone-nodule horizon — and the base of TmB, whose basal lignite yields the *Purussaurus* cranial horizon beneath a winnowed shell bed. Santa Carina spans the top of SJA, the mud-dominated SJB (starved ripples, shell lenses, a silty coarsening-upward top) and the base of SJC. Tamshiyacu is mud- and silt-dominated (86% of the section; no clean sand, no carbonate concretions), whereas Santa Carina is coarser and more fossiliferous (a metre-scale sand body, additional shell beds, and carbonate concretions), consistent with a more basinward bay-centre and a more marginal bay-margin position, respectively, on a low-gradient shoreline.

Geochemistry. XRF shows that total Fe does not track magnetic-mineral concentration (magnetic susceptibility) but instead co-varies with S and Ca+Sr: iron is dominantly authigenic, held in pyrite and carbonate (Fig. 2). Discrete S–Fe–Ca-enriched intervals mark reducing, shell-bearing horizons; intervening beds are more oxidized and detritally-diluted.

DISCUSSION

Age. Four convergent constraints — not all fully independent, since the *Purussaurus* ceiling and the La Venta U–Pb model are interlinked through the Honda Group — point to chron C5AAr, $\sim$13.2–13.4 Ma: (i) the shared reversal and the N–R–N thickness pattern; (ii) a palynological floor (zones T-15/T-16, $>$12.7 Ma; Jaramillo et al., 2011); (iii) a *Purussaurus* maximum-age ceiling ($<$13.8 Ma); and (iv), most decisively, U-Pb detrital-zircon geochronology of the *Purussaurus neivensis*–bearing La Venta section (Honda Group), whose fossil-rich upper La Victoria Formation yields youngest zircon populations of $\sim$13.9 Ma (sample 44011, YSP 13.9$\pm$0.3 Ma) and whose C5AAr interval there spans 13.18–13.36 Ma (Mora-Rojas et al., 2023) — an anchor external to the Pebas. The Tamshiyacu *Purussaurus* cranial horizon lies at 2.71 m, in the TmB basal lignite $\sim$0.17 m above the top of the measured C5AAr record, and is therefore bracketed by C5AAr–C5AAn. At La Venta the dated *P. neivensis* horizon, the Monkey Beds (stratigraphic level 9, $\sim$557–566 m in the composite section), lies $\sim$13 m above the base of C5An.1n in the reinterpreted polarity column and at $\sim$12.0–12.2 Ma in the U-Pb age model (Mora-Rojas et al., 2023) — $\sim$1 Myr younger than Tamshiyacu (Fig. 2). The Iquitos cranial therefore does not share the La Venta chron; it places *Purussaurus* in the Pan-Amazonian wetland $\sim$1 Myr before its best-dated Laventan occurrence, linking the Iquitos Pebas to the older, upper La Victoria part of the Laventan succession rather than to the Monkey Beds. Propagating the U-Pb-calibrated La Venta age model through this magnetostratigraphic tie (Monte Carlo; the model sampled at the correlative La Venta level — the C5AAr top, 176.6 m in the composite section, $-20/+15$ m — and intersected with the chron bracket) yields an inferred age for the Tamshiyacu *Purussaurus* of 13.18 Ma (95% CI 13.04–13.35 Ma; Fig. 5). Notably, the La Venta age model alone is broad at this level ($\sim$13.1 Ma, 95% CI 12.8–13.5 Ma); the sub-million-year precision derives from the C5AAr–C5AAn mag-

netostratigraphic bracket, i.e. from the paleomagnetic correlation rather than the U-Pb dates in isolation. The tie is independently consistent with molluscan biostratigraphy: the hyperdiverse Iquitos crocodylian assemblage, including *Gryposuchus pachakamue*, is assigned to Mollusc Zone 8 ($\sim$13 Ma) in the same sections (Salas-Gismondi et al., 2016), and our C5AAr calibration now places that assemblage on the GPTS to $\sim$180 kyr.

Why C5AAr and not another chron? A single reversal cannot name its own chron: any mid-Miocene reversed chron is a candidate, and the C5AAr assignment is an exclusion argument, made explicit in Fig. 3. Stretching the Santa Carina N–R–N pattern over each candidate, the palynological window (zones T-15/T-16, 12.7–14.2 Ma) eliminates C5Ar.2r and C5Ar.3r (the section top would post-date the pollen floor), and the *Purussaurus* ceiling eliminates C5ABr and C5ACr (the section base would predate the genus). C5AAr is the only surviving placement, but we flag its two open seams: the fit is marginal at the fossil ceiling (implied section base 13.84 Ma), and C5ABr actually matches the polarity-zone thicknesses better under uniform accumulation, failing only the fossil constraint. The detrital-zircon U–Pb maximum depositional ages now in analysis (five samples from Tamshiyacu, Santa Carina, and Ave María) provide the independent, biostratigraphy-free test: a youngest-grain population of $\sim$13.4–13.6 Ma would exclude C5ABr outright, whereas one younger than $\sim$13.2 Ma would falsify C5AAr itself. The same data will anchor the Iquitos sections to the identical U–Pb systematics that calibrate the La Venta composite (Mora-Rojas et al., 2023), so the Pebas–La Venta comparison in Fig. 2 becomes internally consistent rather than merely parallel. *[Working-draft note: a joint TAM–SC boundary fit (in progress) indicates that the C5AAr placement requires $\sim$2.7$\times$ faster net accumulation across Santa Carina's upper normal than across its reversed zone — as the episodic, condensed depositional model predicts — and the incoming detrital-zircon maximum depositional ages, tied into the La Venta U–Pb age model, will refine both the chron assignment and the fossil age.]*

Environment. The authigenic Fe–S–Ca geochemistry records a reducing, sporadically dysaerobic bottom water in which bacterial sulphate reduction pyritized the mud; because sulphate is scarce in freshwater, the punctuated pyrite enrichments most plausibly track brief brackish incursions, and the Ca+Sr shell beds are their winnowed transgressive lags. This independently supports the tidally-influenced, brackish bay-margin interpretation from ichnology and tidal rhythmites at the same localities (Gingras et al., 2002; Räsänen et al., 1995), superimposed on a dominantly freshwater mega-wetland (Hoorn et al., 2010a; Wesselingh & Salo, 2006). The same early sulphate-reducing diagenesis that pyritized these muds also produces the precursor iron sulfide greigite, and the hardest intervals (MDF $\geq$70 mT, with a gyroremanent response above $\sim$70 mT) identify greigite as a likely co-carrier of the remanence alongside magnetite. Because greigite forms during early sulfate reduction, at or near deposition, and because refitting the ChRM without the GRM-affected steps leaves polarity unchanged for 89% of specimens and moves the C5AAr datum by $\leq$1 cm, the polarity chronology is insensitive to the magnetic carrier.

Condensation and correlation. Net accumulation is very low: the C5AAr chron ($\sim$180 kyr) is expressed in only $\sim$0.5 m at Santa Carina ($\sim$3 m/Myr compacted; $\sim$6–9 m/Myr decompacted). This is an order of magnitude below the $\sim$100 m/Myr implied by orbital (obliquity) cyclicity elsewhere in the Pebas (Hoorn et al., 2022) and the $\sim$45–130 m/Myr regional average from the formation's $\sim$350–1000 m thickness (Hoorn et al., 2010a). The discrepancy is diagnostic of *incompleteness*: at a sediment-starved, low-accommodation bay margin (Gingras et al., 2002), rapid intra-cycle deposition was repeatedly interrupted by non-deposition, transgressive erosion (the *Glossifungites* firmgrounds), and winnowing (shell lags), so net accumulation is slow. The same low-accommodation architecture makes lignite flooding surfaces diachronous and explains why the two sections correlate only loosely by lithology; the reversal provides the isochronous tie (Fig. 2), and its offset from the lignite ties quantifies the diachroneity. The regional correlation to the U-Pb-dated La Venta composite section, where C5AAr occupies 153.5–176.6 m, reinforces the C5AAr tie (Fig. 2).

The low accumulation rate is not anomalous (Fig. 4). Across the C5AA interval the Iquitos margin ($\sim$3 m/Myr compacted, $\sim$6–9 decompacted) matches the starved open-marine record at ODP Site 1237 (Nazca

Ridge; ∼7.7 m/Myr; Mix et al., 2003), yet is an order of magnitude below the astronomically tuned Ceara Rise carbonates downstream in the equatorial Atlantic (∼16 m/Myr, ODP Site 926; Backman & Raffi, 1997) and the Pebas basin-scale average (∼45–130 m/Myr; Hoorn et al., 2010a, 2022). A starved, low-accommodation bay margin at ∼13 Ma is therefore expected, and the condensation is a depositional signal, not a correlation artifact.

CONCLUSIONS

Paleomagnetic records from Tamshiyacu and Santa Carina provide the first direct GPTS calibration of the Pebas Formation at Iquitos. Both sections share the C5ABn–C5AAr reversal, and four convergent constraints place the studied interval in chron C5AAr, ∼13.2–13.4 Ma. A *Purussaurus* cranial horizon at Tamshiyacu lies within C5AAr, ∼1 Myr older than the U-Pb-dated La Venta occurrence of *P. neivensis* in the Monkey Beds (C5An.1n). Authigenic Fe–S–Ca geochemistry confirms reducing, episodically brackish conditions consistent with a tidally-influenced bay margin, and very low net accumulation (∼3 m/Myr) identifies the record as condensed and incomplete — the expected signature of a low-accommodation Pan-Amazonian wetland margin. More broadly, by pinning a single fossil horizon to a ∼180 kyr polarity chron and placing it against the U-Pb-dated La Venta section, the approach converts an abundant but coarsely dated fauna into a sub-million-year chronologic framework — resolution suitable for calibrating Neotropical evolutionary and biogeographic models, and a template for dating the many other fossiliferous Pebas cutbanks that at present lack direct GPTS control.

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REFERENCES CITED

Backman, J., and Raffi, I., 1997, Calibration of Miocene nannofossil events to orbitally tuned cyclostratigraphies from Ceara Rise: Proceedings of the Ocean Drilling Program, Scientific Results, v. 154, p. 83–99.

Flynn, J.J., Guerrero, J., and Swisher, C.C., III, 1997, Geochronology of the Honda Group, *in* Kay, R.F., Madden, R.H., Cifelli, R.L., and Flynn, J.J., eds., Vertebrate Paleontology in the Neotropics: The Miocene Fauna of La Venta, Colombia: Washington, D.C., Smithsonian Institution Press, p. 44–59.

Gingras, M.K., Räsänen, M.E., Pemberton, S.G., Romero, L.P., 2002, Ichnology and sedimentology reveal depositional characteristics of bay-margin parasequences in the Miocene Amazonian foreland basin: Journal of Sedimentary Research, v. 72, p. 871–883.

Hilgen, F.J., Lourens, L.J., and Van Dam, J.A., 2012, The Neogene period, *in* Gradstein, F.M., Ogg, J.G., Schmitz, M.D., and Ogg, G.M., eds., The Geologic Time Scale 2012: Amsterdam, Elsevier, p. 923–978.

Hoorn, C., 1993, Marine incursions and the influence of Andean tectonics on the Miocene depositional history of northwestern Amazonia: Palaeogeography, Palaeoclimatology, Palaeoecology, v. 105, p. 267–309.

Hoorn, C., et al., 2010a, Amazonia through time: Andean uplift, climate change, landscape evolution, and biodiversity: Science, v. 330, p. 927–931.

Hoorn, C., Wesselingh, F.P., Hovikoski, J., Guerrero, J., 2010b, The development of the Amazonian mega-wetland (Miocene; Brazil, Colombia, Peru, Bolivia), *in* Hoorn, C., and Wesselingh, F.P., eds., Amazonia: Landscape and Species Evolution: A Look into the Past: Oxford, Wiley-Blackwell, p. 123–142.

Hoorn, C., Kukla, T., Bogotá-Angel, G., et al., 2022, Cyclic sediment deposition by orbital forcing in the Miocene wetland of western Amazonia? New insights from a multidisciplinary approach: Global and Planetary Change, v. 210, 103717.

Jaramillo, C., Rueda, M., Torres, V., 2011, A palynological zonation for the Cenozoic of the Llanos and Llanos Foothills of Colombia: Palynology, v. 35, p. 46–84.

Meade, R.H., 1966, Factors influencing the early stages of the compaction of clays and sands — review: Journal of Sedimentary Petrology, v. 36, p. 1085–1101.

Mix, A.C., Tiedemann, R., Blum, P., et al., 2003, Proceedings of the Ocean Drilling Program, Initial Reports, v. 202: College Station, Texas, Ocean Drilling Program.

Mora-Rojas, L., Cárdenas, A., Jaramillo, C., et al., 2023, Stratigraphy of a middle Miocene neotropical Lagerstätte (La Venta Site, Colombia): Geodiversitas, v. 45(6), p. 197–221.

Räsänen, M.E., Linna, A.M., Santos, J.C.R., Negri, F.R., 1995, Late Miocene tidal deposits in the Amazonian foreland basin: Science, v. 269, p. 386–390.

Salas-Gismondi, R., Flynn, J.J., Baby, P., Tejada-Lara, J.V., Wesselingh, F.P., Antoine, P.-O., 2015, A Miocene hyperdiverse crocodylian community reveals peculiar trophic dynamics in proto-Amazonian mega-wetlands: Proceedings of the Royal Society B, v. 282, 20142490.

Salas-Gismondi, R., Flynn, J.J., Baby, P., Tejada-Lara, J.V., Claude, J., Antoine, P.-O., 2016, A new 13 million year old gavialoid crocodylian from proto-Amazonian mega-wetlands reveals parallel evolutionary trends in skull shape linked to longirostry: PLoS ONE, v. 11, no. 4, e0152453.

Sclater, J.G., and Christie, P.A.F., 1980, Continental stretching: an explanation of the post-mid-Cretaceous subsidence of the central North Sea Basin: Journal of Geophysical Research, v. 85, p. 3711–3739.

Wesselingh, F.P., Salo, J.A., 2006, A Miocene perspective on the evolution of the Amazonian biota: Scripta Geologica, v. 133, p. 439–458.

FIGURE CAPTIONS

Figure 1. Location of the Tamshiyacu (TAM) and Santa Carina (SC) sections along the Amazon River near Iquitos, Peru, with the regional Miocene localities La Venta and Contamana and the approximate extent of the Pebas mega-wetland. Political boundaries and river courses schematic; site markers at true coordinates.

Figure 2. Master magnetostratigraphic correlation. *Left* — the Tamshiyacu (TAM) and Santa Carina (SC) sections, each with lithology (`pystrat` grain-size widths and FGDC swatches; facies after Gingras et al., 2002); a feature lane with grading triangles (fining-/coarsening-upward), *Glossifungites* firmgrounds, bioturbation, root traces, hummocky cross-stratification, ripples, molluscs and concretions (after Gingras et al., 2002 and our field logs); parasequence brackets (TmA–TmB, SJA–SJC) with their basal surfaces (dashed); raw and 7-cm-median ChRM inclination; polarity (hatched = no data, including the NRM-only upper TAM core); and cleaned log-ratio XRF (ln[Fe/coh], ln[S/coh]) over the full scanned interval, all hung on the shared C5AAr reversal (red band, spanning each site's reversed interval). The Tamshiyacu *Purussaurus* (purple) carries its inferred age; the posterior is inset beneath TAM. *Right* — the La Venta composite section (Honda Group; 0–625 m, lithofacies and structures digitised from Mora-Rojas et al., 2023), with the Monkey Beds (StL 9; La Venta *P. neivensis*, dark red) and Fish Bed (StL 11) marked, and its reinterpreted polarity column; the GPTS (ATNTS2012; Hilgen et al., 2012); and the La Venta detrital-zircon U-Pb Bayesian age model (95% CI, labelled with sample height). Shaded ribbons tie C5AAr from Peru to La Venta to the GPTS; dashed lines tie every La Venta reversal to its GPTS boundary (solid dark red: base of C5An.1n, 12.17 Ma, 13 m below the Monkey Beds). Mollusc Zone 8 ($\sim$13 Ma) is shown for regional context.

Figure 3. Candidate GPTS placements for the Santa Carina reversal and the constraints that exclude them. Each row stretches the measured N–R–N section so that its reversed zone occupies one reversed chron of ATNTS2012 (Hilgen et al., 2012); dashed lines mark the palynological window (zones T-15/T-16; Jaramillo et al., 2011) and the *Purussaurus* ceiling, and hatching marks the constraint each placement violates. C5AAr is the only surviving candidate; the detrital-zircon maximum depositional ages in analysis will test it against C5ABr, the best pure-pattern fit.

Figure 4. Middle-Miocene accumulation. *(a)* Age–depth across the C5AA interval (all pinned at the C5AAr base): the Iquitos margin (this study, $\sim$7 m/Myr decompacted) tracks the starved open-marine reference ODP Site 1237 (Nazca Ridge; Mix et al., 2003) and is much slower than the astronomically tuned Ceara Rise carbonate record (ODP Site 926, equatorial Atlantic; Backman & Raffi, 1997). *(b)* Accumulation-rate context on a log scale: the Iquitos site ($\sim$3 m/Myr compacted, 6–9 decompacted) is an order of magnitude below the Pebas basin-scale average ($\sim$45–130 m/Myr; Hoorn et al., 2010a) and the orbital estimate ($\sim$100 m/Myr; Hoorn et al., 2022), identifying it as a genuinely starved sub-environment rather than a miscorrelation artifact.

Figure 5. The Tamshiyacu *Purussaurus* horizon under each candidate chron, *(a)* as measured (compacted) and *(b)* decompacted (mud $\times$2.6; lignite $\times$5). Tamshiyacu preserves only the reversed-over-normal pair — the upper normal was lost in the paramagnetic *Purussaurus*-bearing lignite — so each placement pins the N$\rightarrow$R boundary at the candidate chron base, draws the slowest rate that fits the measured reversed zone inside the chron, and shows the fossil horizon's permitted age bracket (purple; faster rates shift it toward the chron base). Under the surviving C5AAr placement the horizon dates to $\sim$13.1–13.4 Ma, consistent with the Monte Carlo estimate of 13.18 Ma (95% CI 13.04–13.35 Ma). Exclusions carry over from Fig. 3; the detrital-zircon maximum depositional ages, tied into the La Venta U–Pb age model, will tighten both panels.

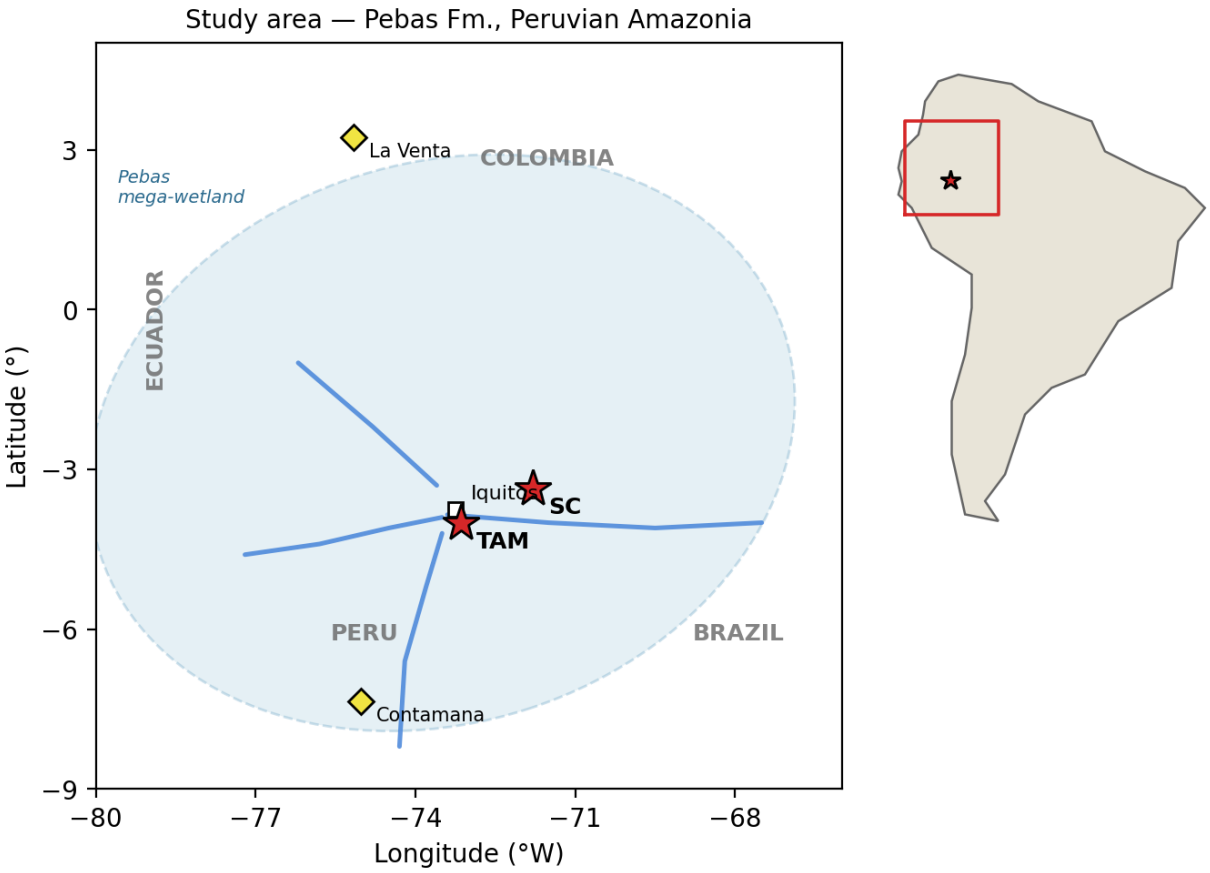
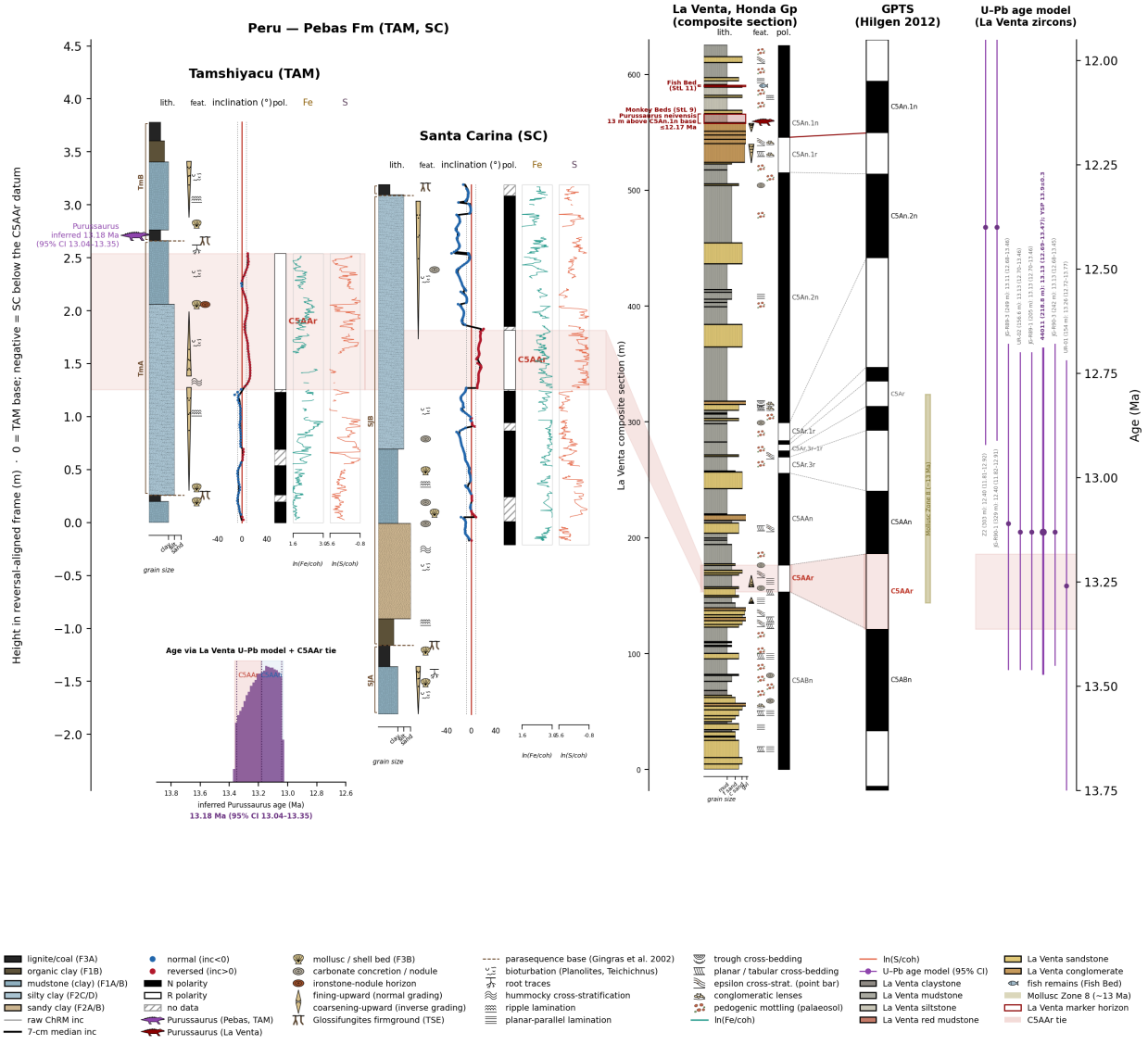
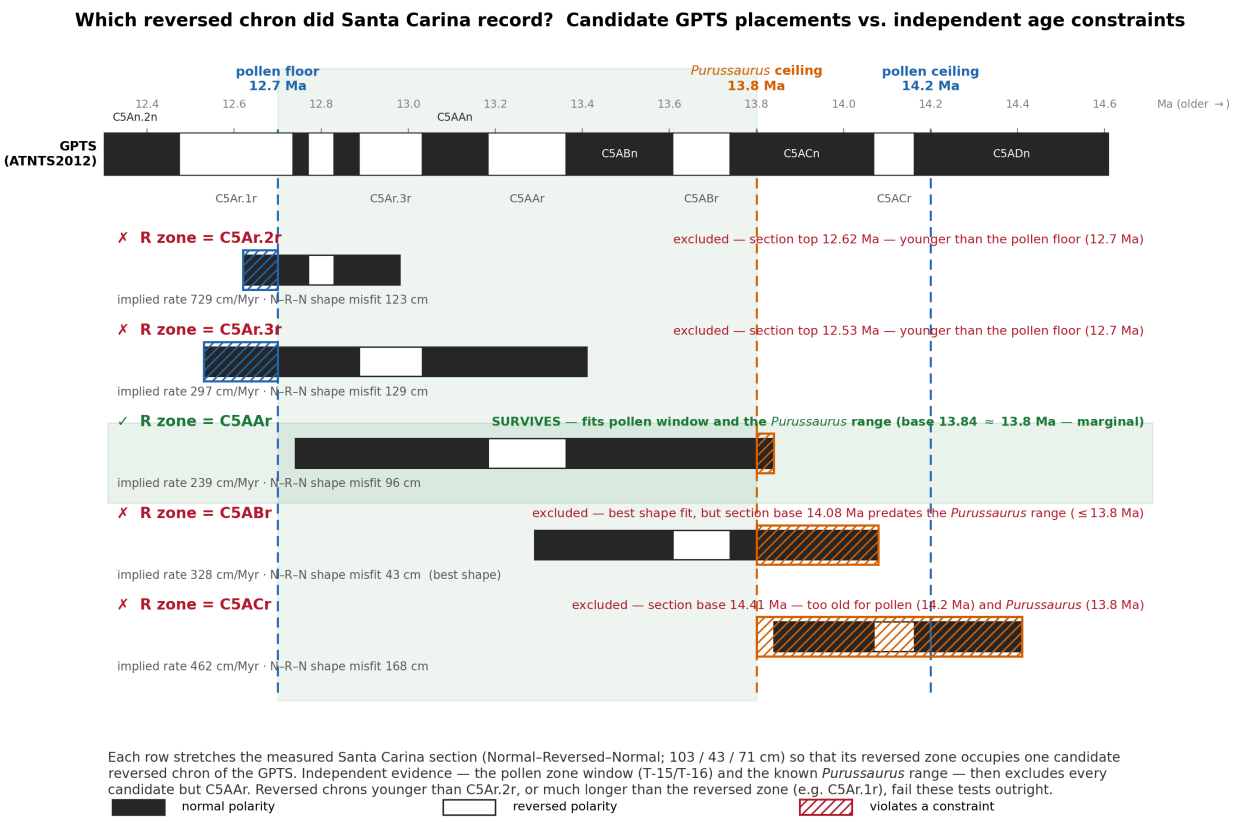

Study area — Pebas Fm., Peruvian Amazonia
La Venta
COLOMBIA
Pebas
mega-wetland
ECUADOR
Iquitos
SC
TAM
PERU
BRAZIL
Contamana
Latitude (°)
Longitude (°W)
3
0
−3
−6
−9
−80
−77
−74
−71
−68

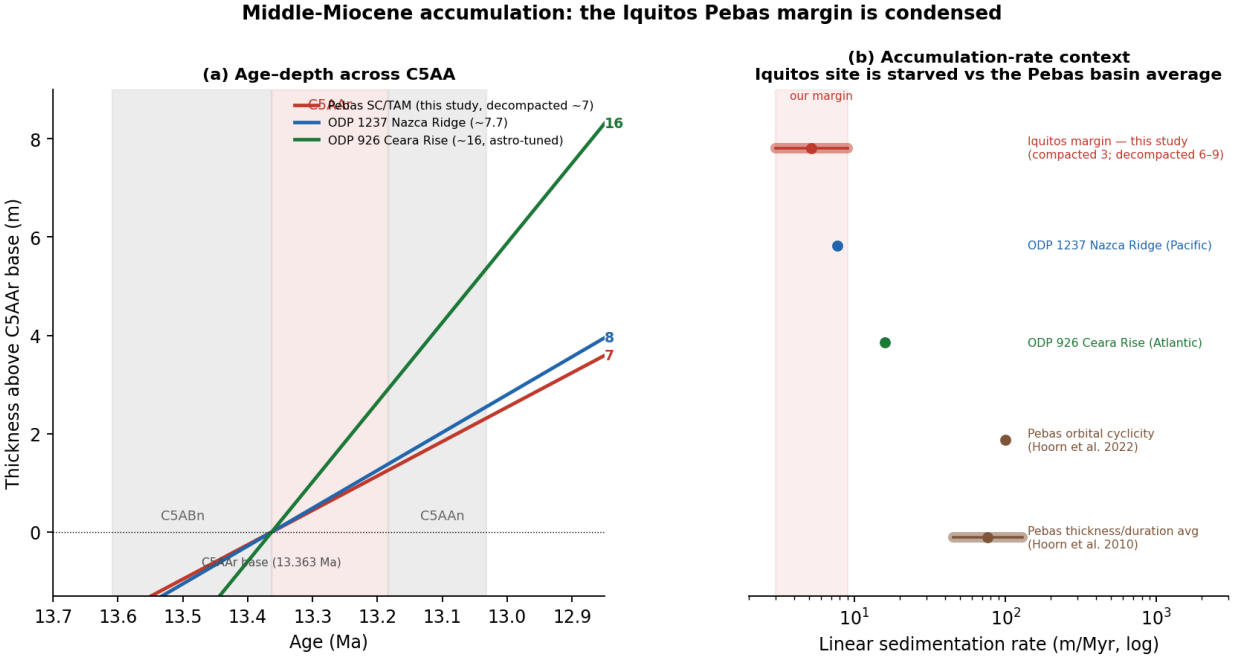
Figure 2. Pebas magnetostratigraphy (TAM, SC) correlated to the U–Pb-calibrated La Venta composite section and the GPTS



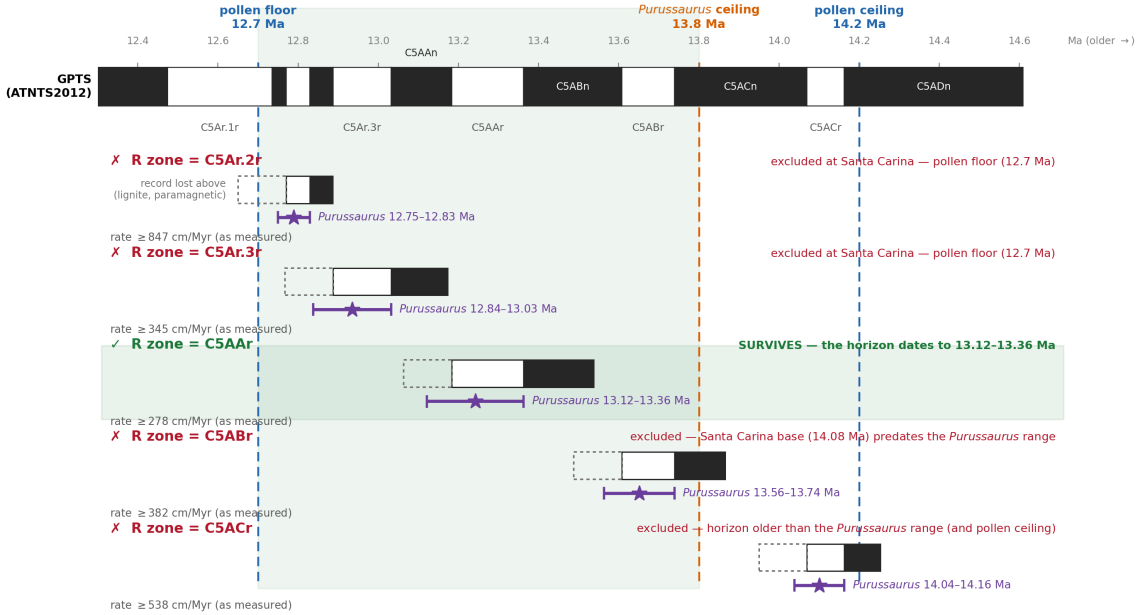
lignite/coal (F3A)
organic clay (F1B)
mudstone (clay) (F1A/B)
silty clay (F2C/D)
sandy clay (F2A/B)
raw ChRM inc
7-cm median inc
normal (inc<0)
reversed (inc>0)
N polarity
R polarity
no data
Purussaurus (Pebas, TAM)
Purussaurus (La Venta)
mollusc / shell bed (F3B)
carbonate concretion / nodule
ironstone-nodule horizon
fining-upward (normal grading)
coarsening-upward (inverse grading)
Glossifungites firmground (TSE)
parasequence base (Gingras et al. 2002)
bioturbation (Planolites, Teichichnus)
root traces
hummocky cross-stratification
ripple lamination
planar-parallel lamination
trough cross-bedding
planar / tabular cross-bedding
epsilon cross-strat. (point bar)
conglomeratic lenses
pedogenic mottling (palaeosol)
ln(Fe/coh)
ln(S/coh)
U–Pb age model (95% CI)
La Venta claystone
La Venta mudstone
La Venta siltstone
La Venta red mudstone
La Venta sandstone
La Venta conglomerate
fish remains (Fish Bed)
Mollusc Zone 8 (~13 Ma)
La Venta marker horizon
C5AAr tie

Lithology after pystrat (grain-size widths + FGDC swatches); TAM & SC parasequences, grading, ichnology & physical structures after Gingras et al. 2002 (Tamshiyacu TmA/TmB; Santa Carina = their Santa Julia SJA-SJC; tied on the lignite-capped flooding surfaces); La Venta composite section (lithology, lithofacies codes, grading) digitised from Mora-Rojas et al. 2023 suppl. s6, polarity from their s15 reinterpretation of Flynn et al. 1997, reversals tied to Hilgen et al. 2012; Peru columns aligned on the shared C5AAr reversal; XRF cleaned (cps≥80k, MSE≤3.5, gaps unbridged, 3-pt median) and shown as log-ratios to coherent scatter (Weltje & Tjallingii 2008); Monkey Beds (Stl. 9) & Fish Bed (Stl. 11) from Mora-Rojas et al. 2023 (suppl. s6, s10) with U–Pb Bayesian age model (detrital zircons, 95% CI); La Venta Purussaurus (P. neivensis, Monkey Beds) after Carrillo et al. 2023; Mollusc Zone 8 after Wesselingh et al. 2006 & Salas-Gismondi et al. 2016. DRAFT.


Which reversed chron did Santa Carina record? Candidate GPTS placements vs. independent age constraints
pollen floor
12.7 Ma
Purussaurus ceiling
13.8 Ma
pollen ceiling
14.2 Ma
12.4
12.6
12.8
13.0
13.2
13.4
13.6
13.8
14.0
14.2
14.4
14.6
Ma (older →)
C5An.2n
C5AAn
GPTS
(ATNTS2012)
C5ABn
C5ACn
C5ADn
C5Ar.1r
C5Ar.3r
C5AAr
C5ABr
C5ACr
✗ R zone = C5Ar.2r
excluded — section top 12.62 Ma — younger than the pollen floor (12.7 Ma)
implied rate 729 cm/Myr · N–R–N shape misfit 123 cm
✗ R zone = C5Ar.3r
excluded — section top 12.53 Ma — younger than the pollen floor (12.7 Ma)
implied rate 297 cm/Myr · N–R–N shape misfit 129 cm
✓ R zone = C5AAr
SURVIVES — fits pollen window and the Purussaurus range (base 13.84 ≈ 13.8 Ma — marginal)
implied rate 239 cm/Myr · N–R–N shape misfit 96 cm
✗ R zone = C5ABr
excluded — best shape fit, but section base 14.08 Ma predates the Purussaurus range (≤ 13.8 Ma)
implied rate 328 cm/Myr · N–R–N shape misfit 43 cm (best shape)
✗ R zone = C5ACr
excluded — section base 14.41 Ma — too old for pollen (14.2 Ma) and Purussaurus (13.8 Ma)
implied rate 462 cm/Myr · N–R–N shape misfit 168 cm
Each row stretches the measured Santa Carina section (Normal–Reversed–Normal; 103 / 43 / 71 cm) so that its reversed zone occupies one candidate reversed chron of the GPTS. Independent evidence — the pollen zone window (T-15/T-16) and the known Purussaurus range — then excludes every candidate but C5AAr. Reversed chrons younger than C5Ar.1r, or much longer than the reversed zone (e.g. C5Ar.1r), fail these tests outright.
normal polarity
reversed polarity
violates a constraint


Middle-Miocene accumulation: the Iquitos Pebas margin is condensed
(a) Age–depth across C5AA
C5AAr
Pebas SC/TAM (this study, decompacted ~7)
ODP 1237 Nazca Ridge (~7.7)
ODP 926 Ceara Rise (~16, astro-tuned)
16
8
7
C5ABn
C5AAn
C5AAr base (13.363 Ma)
Thickness above C5AAr base (m)
Age (Ma)
13.7
13.6
13.5
13.4
13.3
13.2
13.1
13.0
12.9
0
2
4
6
8
(b) Accumulation-rate context
Iquitos site is starved vs the Pebas basin average
our margin
Iquitos margin — this study
(compacted 3; decompacted 6–9)
ODP 1237 Nazca Ridge (Pacific)
ODP 926 Ceara Rise (Atlantic)
Pebas orbital cyclicity
(Hoorn et al. 2022)
Pebas thickness/duration avg
(Hoorn et al. 2010)
Linear sedimentation rate (m/Myr, log)
10^1
10^2
10^3

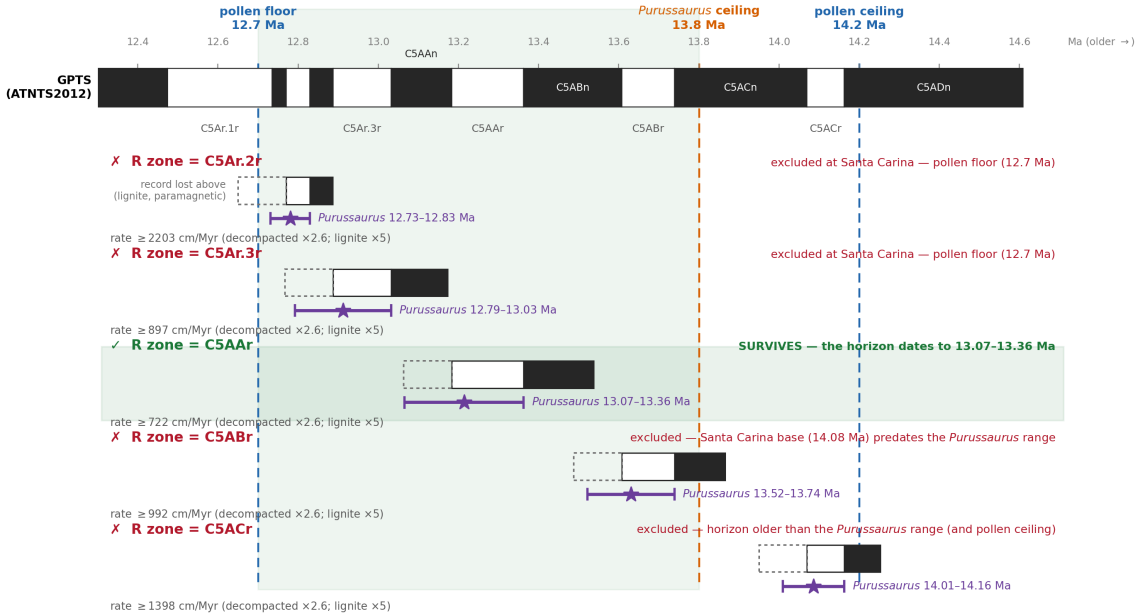
Tamshiyacu: the *Purussaurus* horizon under each candidate chron — as measured (compacted)



Tamshiyacu preserves only R-over-N: the upper normal was lost in the paramagnetic *Purussaurus*-bearing lignite (Group 1), so each row pins the N→R boundary at the candidate chron base and draws the section at the slowest rate that fits the measured reversed zone (50 cm) inside the chron; faster rates shift the horizon (star) toward the chron base. Exclusions carry over from the Santa Carina N–R–N test (Fig. 3).

normal polarity reversed polarity ★ *Purussaurus* cranial horizon (age bracket)

Tamshiyacu: the *Purussaurus* horizon under each candidate chron — decompacted (mud ×2.6, lignite ×5)



Tamshiyacu preserves only R-over-N: the upper normal was lost in the paramagnetic *Purussaurus*-bearing lignite (Group 1), so each row pins the N→R boundary at the candidate chron base and draws the section at the slowest rate that fits the measured reversed zone (50 cm) inside the chron; faster rates shift the horizon (star) toward the chron base. Exclusions carry over from the Santa Carina N–R–N test (Fig. 3).

normal polarity reversed polarity ★ *Purussaurus* cranial horizon (age bracket)